

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&E / E&C) (Credit System) Degree Examinations

April 2023

21EE301 / 21EC301 – VECTOR CALCULUS AND TRANSFORM TECHNIQUES

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) A particle moves along the curve $x = t^3 + 1$, $y = t^2$, $z = 2t + 3$ where t is the time. Find the components of its velocity and acceleration at $t = 1$ in the direction $\hat{i} + \hat{j} + 3\hat{k}$.	6	L*1	1	1
	b) Prove that cylindrical coordinate system is an orthogonal curvilinear coordinate system.	7	L2	1	2
	c) Using Stoke's theorem evaluate $\int_C [(x+y)dx + (2x-z)dy + (y+z)dz]$ where C is the boundary of the triangle with vertices $(2, 0, 0)$, $(0, 3, 0)$ & $(0, 0, 6)$.	7	L2	2	2
2.	a) Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction of the vector $2\hat{i} - \hat{j} - 2\hat{k}$.	6	L1	1	1
	b) State and prove Green's theorem.	7	L2	2	2
	c) If $\vec{F} = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$ when C is made of the straight-line segments from $(0, 0, 0)$ to $(1, 0, 0)$ from $(1, 0, 0)$ to $(1, 1, 0)$ and then from $(1, 1, 0)$ to $(1, 1, 1)$.	7	L2	2	2
3.	a) Prove that $\vec{V} = (y^2 - z^2 + 3yz - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$ is both solenoidal and irrotational.	6	L1	1	1
	b) Prove that $\int_C \vec{F} \cdot d\vec{r}$ is independent of the path of integration if and only if $\oint_C \vec{F} \cdot d\vec{r} = 0$, where C is any simple closed curve.	7	L2	2	2
	c) Evaluate using Gauss divergence theorem $\iiint_S \vec{F} \cdot \hat{n} ds$ where $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ and S is the surface bounding the region $x^2 + y^2 = 4$, $z = 0$ & $z = 3$.	7	L2	2	2
Unit – II					
4.	a) Determine the imaginary part of the analytical function $f(z)$ whose real part is $u = x^2 - y^2 - y$.	6	L1	3	1
	b) If $f(z) = u(x, y) + i v(x, y)$ is analytic in a region R , then prove that $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ & $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.	7	L2	3	2
	c) Evaluate $\int_C \frac{e^{2z}}{(z-1)(z-2)} dz$, if $c: z = 3$ using Cauchy's integral formula.	7	L2	4	2
5.	a) Find the poles and residues at each pole of the function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$.	6	L1	4	1
	b) State and prove Cauchy's integral theorem.	7	L2	4	2

- c) Find the bilinear transformation which maps $z = 1, 0, -1$ onto $w = i, 1, \infty$. 7 L2 3 2
6. a) Show that $w = \log z$ is analytic and hence find $\frac{dw}{dz}$. 6 L1 3 1
- b) Write the Laurent's series expansion of $\frac{1}{z^2 - 3z + 2}$ in the region $1 < |z| < 2$. 7 L2 4 2
- c) Evaluate $\int_C \frac{z-1}{(z+1)^2(z-2)} dz$ where C is the circle $|z - i| = 2$ using residue theorem. 7 L2 4 2

Unit – III

7. a) Find the Fourier series of $f(x) = \begin{cases} -k, & (-\pi, 0) \\ k, & (0, \pi) \end{cases}; f(x + 2\pi) = f(x)$. 6 L1 5 1
- b) Find the half range cosine series of $f(x) = x^2$ in $(0, \pi)$. 7 L2 5 2
- c) Find a_0, a_1, b_1 of the Fourier series of the following data.
- | | | | | | | | |
|---|-----|---------|----------|-------|----------|----------|--------|
| x | 0 | $\pi/3$ | $2\pi/3$ | π | $4\pi/3$ | $5\pi/3$ | 2π |
| y | 1.0 | 1.4 | 1.9 | 1.7 | 1.5 | 1.2 | 1.0 |
- 7 L2 5 2
8. a) Find the z- transform and region of convergence of $u_n = \begin{cases} 5^{-n}, & \text{if } n \geq 0 \\ k, & \text{if } n < 0 \end{cases}$. 6 L1 5 1
- b) State and prove convolution theorem of bilateral Z- transform. 7 L2 5 2
- c) Using Z- transforms solve the following difference equation. $u_{n+2} + 4u_{n+1} + 3u_n = 3^n, u_0 = 0, u_1 = 1$. 7 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
